

Supplementary Information

The behavioural-equivalence manifold of whole-organism nervous systems

Chenxi He, Cavendish Laboratory, University of Cambridge
ch2067@cam.ac.uk

Contents

1	Measuring the manifold: methods	2
1.1	Multistart parameter drift (S1.1)	2
1.2	Hessian eigenspectrum (S1.2)	2
1.3	Simulation-based calibration as a manifold probe (S1.3)	3
1.4	B1 — per-mechanism stiff/sloppy (S1.4)	3
2	External references (no simulator used)	3
2.1	E1 — Prinz–Marder pyloric population (S2.1)	3
2.2	E2 — real neuron variability (S2.2)	4
2.3	E4 — ion-channel-gene conservation (S2.3)	4
2.4	E5 — isogenic behavioural manifold (S2.4)	4
3	Unified same-region probe test (result)	4
4	Confirmatory experiments	4
4.1	Higher-dimensional Hessian: FlyGym 48-d (S5.1)	4
4.2	Per-channel stiff/sloppy decomposition (S5.2)	5
4.3	Full-connectome manifold probe (S5.3)	5
4.4	Behaviour-based prediction of the internal neural state (S5.4)	5
4.5	Mutant mechanism localisation (S5.5)	5
4.6	External mutant-phenotype check (S5.6, null)	6
4.7	A fifth simulator: <i>Drosophila</i> connectome model (S5.7)	6
4.8	Cross-system stiff/sloppy convergence (S5.8)	7
5	The identifiable subspace: identity, tiling, and cross-species universality	8
5.1	Identification of the manifold with the eigenworm postural space (S6.1)	8
5.2	Independence of the low dimension from observable resolution (S6.2)	8
5.3	Complementary tiling: distinct behaviours constrain distinct axes (S6.3)	8
5.4	Union growth and saturation within one model (S6.4)	8
5.5	Cross-species: the same growth structure across three phyla (S6.5)	9
5.6	Concordance of two independently constructed worm models (S6.6)	9
5.7	Sloppy directions as coupled parameter combinations (S6.7)	9
5.8	Robustness of the effective-dimension measurement (S6.8)	10
5.9	Robustness of the low dimension to the spectral-mass threshold (S6.9)	10

This Supplementary Information provides the extended methods and the complete per-experiment quantities underlying each main-text claim. It comprises: the measurement methods and their controls; the provenance of the external biological datasets; the confirmatory experiments that extend the probes to higher parameter dimension, to named single channels, to the full connectome, to the internal neural state, and to a fifth simulator; and the experiments that identify the manifold with the eigenworm space, establish its complementary tiling by behaviour, and demonstrate the structure across three phyla, together with one inconclusive control that is reported in full. Every quantity is drawn from a saved result file, named per section, and is regenerated from that file by the accompanying reproduction code.

1 Measuring the manifold: methods

1.1 Multistart parameter drift (S1.1)

For each simulator we run N independent optimisations (separable CMA-ES or SPSA) from independent starts against one *fixed* behavioural target, and record (i) behavioural recovery (fraction of the random-to-target behaviour-distance closed), (ii) the normalised per-dimension dispersion of the N recovered parameter vectors, and (iii) the PCA participation ratio of the solution cloud. A behaviourally-converged but parameter-dispersed cloud is direct evidence of a positive-dimensional equivalence manifold.

Table 1: Fixed-target multistart. “behav. rec.” is the median behavioural recovery; “ θ disp.” the normalised per-dimension dispersion across the N solutions; “cloud dim” the PCA participation ratio.

sim / method	dim	N	success	behav. rec.	θ disp.	cloud dim
FlyGym / CMA-ES	126	30	29/30	70.8%	0.098	21.7/126
BAAIWorm / SPSA	5	10	5–7/10	—	0.67	—

The FlyGym 126-d cloud occupies an effective 21.7 of 126 dimensions. (The flybody 515-d objective overfits per-seed without a prior and is excluded from the drift panel; it appears only in the Hessian analysis.) On the full coupled **real modWorm** model (OpenWorm connectome, 279 neurons), the manifold is read from the per-mechanism Hessian (S1.2): only 2 of 7 biophysical-mechanism directions carry behavioural curvature, the rest forming the flat manifold. (A direct random-perturbation probe gives the same qualitative picture but with an assay-dependent magnitude, so we report the manifold by its effective dimension, not by the raw sloppiness ratio, which is not comparable across simulators with different observables.) File `modworm_REAL_manifold.json`.

1.2 Hessian eigenspectrum (S1.2)

At each converged optimum we form the behavioural-loss Hessian by finite differences (low dimension) or a Lanczos approximation (high dimension) and report the effective dimension (number of eigendirections carrying 90%/99% of the spectral mass) and the count of near-flat directions ($|\lambda|$ at the finite-difference floor).

- **BAAIWorm, 5 mechanism-class scalings** ($\varepsilon=5 \times 10^{-3}$): eigenvalues $\lambda_1=+348$, $\lambda_2=-340$, $\lambda_3=-214$ (saddle), $\lambda_4 \approx 0$ (ion), $\lambda_5 \approx 0$ (passive); the ion and passive eigendirections are

flat to machine precision over a parameter L2 length of 2.0. Sign structure stable across $\varepsilon \in [10^{-4}, 5 \times 10^{-2}]$.

- **flybody, 767-parameter policy** (Lanczos $m=40$, $\varepsilon=0.005$, anchor $L_0=9.513$): six positive stiff eigenvalues (8.2×10^4 down to 2.1×10^3); 28/40 sampled directions flat ($|\lambda| \sim 10^{-11}$); effective dimension 6 at 90%, 11 at 99%; participation ratio 3.76.
- **modWorm, per-mechanism Gauss–Newton Hessian** (full coupled neural+body model on the OpenWorm connectome (1, 2), 279 neurons; behavioural elasticity over 7 biophysical mechanisms): effective dimension 2 of 7 at both 90% and 99% of spectral mass; participation ratio 1.29. The stiff mechanisms are the synaptic and gap-junction conductances and the synaptic rise time; the membrane capacitance and the sigmoid gain are sloppy (zero elasticity)—again indicating the primacy of connectivity over single-cell intrinsics. File `modworm_REAL_b1.json`.

1.3 Simulation-based calibration as a manifold probe (S1.3)

An amortised neural posterior (masked autoregressive flow (3), `sbi` (4)) is trained once on the prior, and per-dimension rank-based SBC (5, 6) is run (M/L posterior/rank draws); “pass” is a per-axis Kolmogorov–Smirnov test at $\alpha=0.05$. Pass counts: real modWorm 7/7 ($N=300$ coupled-model sims), FlyGym 22/48, BAAIWorm full5 5/5. FlyGym posterior-mean gain rel-error 0.018. The partial pass on FlyGym delimits the calibratable subspace of the manifold; the full pass on the lower-dimensional real-modWorm and BAAIWorm probes reflects a correctly recovered posterior.

1.4 B1 — per-mechanism stiff/sloppy (S1.4)

A preliminary probe scaled five global mechanism classes and evaluated the raw finite-difference Hessian rather than its Gauss–Newton form. The per-parameter curvature magnitudes ordered the classes as synaptic 331.97, gap-junction 299.64, motor-output 270.50 (stiff) against ion-channel and passive-membrane ≈ 0 (sloppy), but the raw Hessian at that point carried mixed-sign eigenvalues (+348, −340, −214, 0, 0), indicating a saddle rather than a minimum. Curvature signs are therefore not interpretable at that point, and we do not use this probe for any reported result. It is superseded by the positive semi-definite Gauss–Newton decomposition over 20 named mechanisms (§5.2), which resolves the same classes at finer granularity, and by the Gauss–Newton analyses of every other system reported here. We retain it only to document the ordering it produced, which agrees with the Gauss–Newton result. File: `zerocompute_crosscheck.json`.

2 External references (no simulator used)

2.1 E1 — Prinz–Marder pyloric population (S2.1)

We re-gate the published conductance-based STG model population (7) to the canonical triphasic pyloric rhythm (all three cells bursting; period $\in [500, 2000]$ ms; strict AB/PD<LP<PY phase ordering; valid rate 0.56%) and probe valid points with the *same* Hessian-eigenspectrum method as S1.2, providing a direct comparison to B1. Over $K=8$ valid points: participation effective dimension 2.13/31; spectrum spanning ~ 10 orders of magnitude; 2–6 stiff directions; stiff = leak and inhibitory-synapse conductances, sloppy = fast Na conductances, consistent with the compensation and co-regulation documented for this circuit (8–11). Files: `e1b_results/e1b_hessian.json`, `e1b_results/e1b_gate.json`.

2.2 E2 — real neuron variability (S2.2)

Allen Cell Types electrophysiology, $n=1914$ mouse+human cortical neurons: per-property coefficient of variation 0.08–1.85 (stiff: resting potential 0.08, fast-trough 0.12; sloppy: adaptation 1.85, firing rate 1.11); property-space PCA participation dimension 4.78/11. File: `E2_allen_variability.json`.

2.3 E4 — ion-channel-gene conservation (S2.3)

40 *C. elegans* ion-channel/synaptic genes, mean ortholog %identity across *Caenorhabditis* (WormBase ParaSite / Ensembl Metazoa): range 77.94–99.91 (median 91.20); most conserved `unc-9` (gap junction, 99.91), `egl-19` (L-type Ca, 97.19); least `unc-49` (GABA-A, 77.94). File: `E4_channel_conservation.json`.

2.4 E5 — isogenic behavioural manifold (S2.4)

OpenWorm Movement Database, $n=61$ isogenic N2 worms, 683 Tierpsy features: cross-individual participation dimension 9.75; 24 PCs for 90% variance, 50 for 99% (PC1 26.6%, PC2 10.7%). File: `E5_n2_manifold.json`.

3 Unified same-region probe test (result)

Computing the probes on the full coupled modWorm model in one shared prior region (per-mechanism log-multipliers, prior scale 0.15). **The two geometric probes agree:** per-mechanism Hessian stiffness versus negative manifold drift give Spearman $\rho=0.96$ ($p=5 \times 10^{-4}$; 7 mechanisms). **The amortised posterior is a calibration control, not a third ranking:** an NPE/MAF (`sbi`) trained on $N=300$ real modWorm simulations is well-calibrated on all 7 mechanisms (per-axis SBC 7/7, Kolmogorov–Smirnov $p > 0.05$), confirming the inference is sound; because a calibrated posterior calibrates every axis, SBC serves as a control rather than a per-axis identifiability ranking. We therefore report a two-probe geometric agreement plus a calibration control. Files: `modworm-REAL_sbc.json`, `modworm-REAL_b1.json`.

4 Confirmatory experiments

These measurements extend the picture above to higher parameter dimension, to named single channels, to the full connectome, and to the internal neural state. Behavioural curvature is reported as the Gauss–Newton Hessian $H=J^\top W J$ formed from the observable Jacobian $J_{ik}=\partial \text{obs}_i / \partial \theta_k$ (whitened by per-observable scale W), the Fisher-information form that quantifies which parameter directions behaviour constrains.

4.1 Higher-dimensional Hessian: FlyGym 48-d (S5.1)

The behavioural Hessian over the 48 actuator gains of the FlyGym walking controller has effective dimension 3 (90%) / 6 (99%) of 48 (participation ratio 2.15), with an eigenspectrum spanning 19.5 orders of magnitude (leading eigenvalues 20.4, 5.5, 2.7, 1.2, 0.80, 0.62, then < 0.06). A controller of substantially higher parameter dimension is nonetheless governed by a 3–6-dimensional stiff subspace, consistent with the low dimensionality measured at the 5-, 30- and 767-parameter scales. File `flygym_hessian48.json`.

4.2 Per-channel stiff/sloppy decomposition (S5.2)

Replacing the five global mechanism-class scalings with 20 named mechanisms—16 individual ion-channel conductances, the passive leak, the synaptic and gap-junction weights, and the motor output—the per-channel behavioural elasticity ($d \ln \text{obs} / d \ln \theta$) has effective dimension 2 (90%) / 3 (99%) of 20 (eigenvalues 43.2, 5.84, 0.88, 0.47, remainder ≈ 0). The stiff mechanisms are the passive leak, the gap-junction and synaptic weights, the motor output, and a few potassium conductances (EGL-36, EGL-2, IRK); the sloppiest are the fast voltage-gated calcium channels EGL-19 (elasticity 0.49) and UNC-2 (0.31) and the calcium-activated potassium channels SLO-1/SLO-2 (0.01–0.16). Behaviour is set by wiring, passive properties and a subset of potassium channels, and is nearly invariant to the fast calcium and calcium-activated potassium conductances—the conductances known to be compensated in small circuits (12). The infinitesimal five-mechanism curvature of §1.4 and this finite per-channel elasticity are distinct measures (an infinitesimal whole-class scaling versus a per-channel finite elasticity); the elasticity is the behaviourally meaningful one and is reported as primary. File `baai_perchannel_hessian.json`.

4.3 Full-connectome manifold probe (S5.3)

Over the full 3076 connection weights (1992 synaptic + 1084 gap-junction), we compare the behavioural change along random directions against matched-norm coherent (block-coordinated) directions. Random directions perturb behaviour as much as or more than coherent ones (coherent/random behavioural-distance ratio 0.57 at per-connection log-scale 0.08, 0.84 at 0.25) and essentially no direction is flat (flat fraction ≈ 0). The connection-weight space is therefore collectively stiff: the behavioural-equivalence manifold’s sloppy directions lie in the biophysical channel parameters (§5.2), not in the synaptic and gap-junction weights. This is consistent with the gap-junction and synaptic axes being stiff and with the robustness of circuit output to channel variation (7, 12). Files `deposit_baai_connectome_manifold_sigma{008,025}.json`.

4.4 Behaviour-based prediction of the internal neural state (S5.4)

Rolling out BAAIWorm (NEURON (13)) with behaviour-recovered parameters and reading the internal membrane voltage of all 136 neurons, the generated neural dynamics are low-dimensional (PCA participation dimension ≈ 4.7 –4.8) with mean absolute pairwise correlation 0.37–0.39—the same regime as the real whole-brain calcium recordings of Kato et al. (14) (mean absolute correlation 0.21). Behaviour-only parameter generation thus reproduces the gross statistics of the unobserved neural state, though the recovered and default arms are similar and the comparison is qualitative (membrane voltage versus calcium fluorescence; behaving versus immobilised). Files `_b2_stats_*.json`.

4.5 Mutant mechanism localisation (S5.5)

Knocking each named mechanism to $0.4\times$ and asking whether the resulting behavioural-shift signature identifies the perturbed mechanism: the sloppy calcium channels are behaviourally near-silent (signature norm EGL-19 0.07, UNC-2 0.12, SLO-1/SLO-2 0.09–0.16) and so cannot be localised from behaviour, while the detectable mechanisms (synaptic, passive, motor) are mutually confusable (cosine > 0.9 with several others). Behaviour alone does not uniquely localise the perturbed mechanism—the mechanism-level expression of the behavioural-equivalence manifold. File `baai_b3_localise.json`.

pattern generator constrains few mechanisms (low-dimensional), whereas a precision visual code constrains many (high-dimensional). The unifying invariant is which axes are stiff: in every case the stiff axes are the behaviourally load-bearing mechanisms (here the T4/T5 motion-detector pathway), and the sloppy axes are the auxiliary ones (input gain at the photoreceptors). The low-dimensional manifold is thus a property of whole-organism motor behaviour specifically, not a universal claim about all neural computations. (We compare simulators by effective dimension rather than by a raw perturbation-sensitivity ratio: across the four motor models the latter ranges over three orders of magnitude with the observable and parameterisation, whereas the Hessian effective dimension is consistently low.) Files `flyvis_b1.json`, `flyvis_manifold.json`.

4.8 Cross-system stiff/sloppy convergence (S5.8)

Applying the same Hessian-eigenspectrum probe to the canonical Prinz–Marder stomatogastric conductance population (a non-worm, non-fly crustacean circuit; §2.1) gives effective dimension 2.13/31 with stiff = leak and inhibitory-synapse conductances and sloppy = fast voltage-gated Na and delayed-rectifier K—the same biophysical split seen in the worm models, where the fast voltage-gated calcium channels (EGL-19, UNC-2) are sloppy (§5.2). Across four real models and three phyla, behaviour/output is set by network wiring and slow/passive properties and left free along the fast voltage-gated inward currents. File `STG_b1analog.json`.

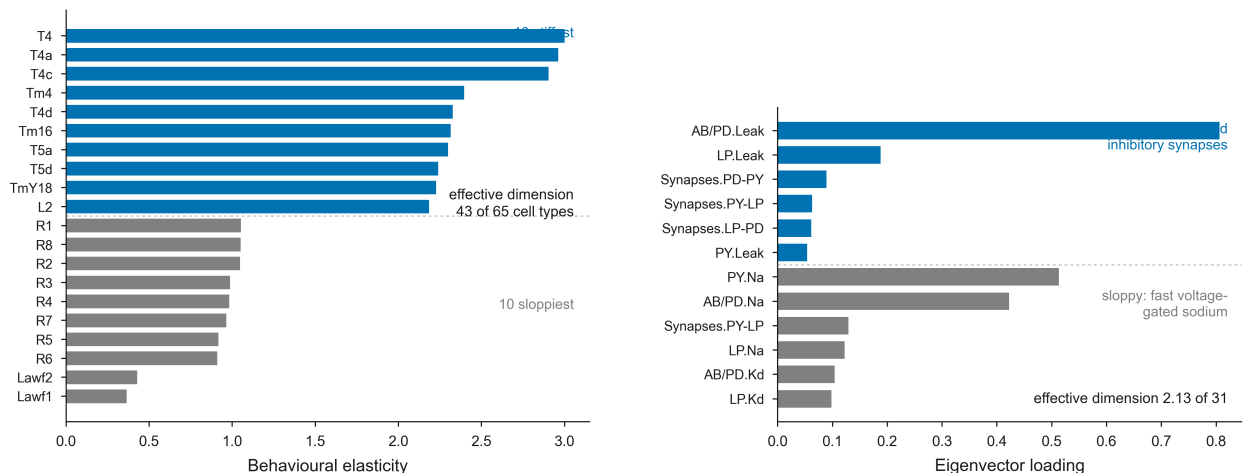


Figure 2: extbfStiff and sloppy structure in a connectome-constrained visual model and in the crustacean pyloric population. extbfa Behavioural elasticity of the flyvis optic-lobe model across its 65 named cell types, showing the ten stiffest above the dashed line and the ten sloppiest below. Blue: stiff; grey: sloppy. The elementary motion detectors T4a–d and their medulla inputs are stiff, whereas photoreceptors and lamina wide-field cells are sloppy, and the effective dimension is 43 of 65. extbfb Eigenvector loadings of the rhythm-gated Prinz–Marder pyloric population, with the six largest stiff loadings above the dashed line and the six largest sloppy loadings below. Stiff, leak and inhibitory-synapse conductances; sloppy, fast voltage-gated sodium; effective dimension 2.13 of 31 over $K=8$ valid points. Note that the biophysical split is the same in both systems and in the worm models of the main text (connectivity and slow or passive properties stiff, fast voltage-gated inward currents sloppy), while the effective dimension is not: a precision visual code constrains many more directions than a whole-body motor pattern. Both probes use the identical Gauss–Newton method and the same finite-difference step.

5 The identifiable subspace: identity, tiling, and cross-species universality

5.1 Identification of the manifold with the eigenworm postural space (S6.1)

We reduce real *C. elegans* skeletons (OpenWorm Movement Database, N2; 8,432 frames) and model body postures to tangent-angle eigenmodes by PCA. Real posture has effective dimension 3.79 (49-point skeletons) / 4.14 (17-point), with four eigenworms capturing 0.96 / 0.94 of postural variance—reproducing the classical four-eigenworm decomposition (18). Two independently built worm models produce behaviour that lives in this same real eigenworm space: the modWorm posture subspace aligns with the real eigenworm basis at mean principal-angle $\cos 0.773$, and BAAIWorm at $\cos 0.68\text{--}0.69$ (67–72% of variance). Within each model it is the stiff mechanisms (synaptic, gap-junction, synaptic-rise/fall time) that move the eigenworm trajectory and the sloppy ones that do not (stiff/sloppy eigenworm-displacement ratio 1.17). The low-dimensional behavioural-equivalence manifold is therefore the neural control of the worm’s ~ 4 -dimensional posture. These alignments are significant against a null of random four-dimensional subspaces: drawing random $k=4$ subspaces in a d -dimensional posture ambient and computing the same mean principal-angle cosine, the modWorm value 0.773 has $p=0.0014$ and BAAIWorm 0.683 has $p=0.043$ even in a deliberately conservative $d=10$ ambient, falling to $p<10^{-4}$ at $d=20$ and $d=48$ (the full tangent-angle space); the random-subspace mean is 0.55/0.38/0.24 at $d=10/20/48$ (20,000 draws each). Files `EW_eigenworm.json`, `EW_baai_local.json`, `NCS_p1p5.json`.

5.2 Independence of the low dimension from observable resolution (S6.2)

Re-measuring the real modWorm behavioural Hessian with 7,200–14,400 posture observables (full body-segment angle and curvature time series, rather than six summary kinematics) leaves the effective dimension at 2 (90%) / 3 (99%) of 7, with spectrum $[1, 0.16, 0.013, 0.002, 0, \dots]$. The low dimensionality is a property of the behaviour, not of how coarsely it is sampled. File `RICH_observable.json`.

5.3 Complementary tiling: distinct behaviours constrain distinct axes (S6.3)

In BAAIWorm we contrast two behaviours—food-directed chemotaxis (food close; base net displacement 0.061) and plain locomotion in a weak gradient (food far; 0.231)—and perturb either the chemosensory parameters or the motor parameters. The chemosensory parameters are stiff during chemotaxis (behavioural sensitivity 0.872) but sloppy during locomotion (0.122): a $7.15\times$ context shift. The motor parameters show the opposite contrast (0.109 chemotaxis vs 0.357 locomotion, ratio 0.30). Each behaviour makes stiff the axis its function needs, and the two behaviours make *different* axes stiff—so the identifiable subspace is the union over behaviours, not the few directions of any one. File `BAAI_chemo.json`.

5.4 Union growth and saturation within one model (S6.4)

On the real modWorm model, the union effective dimension over a behaviour set grows with the number of behaviours and then saturates: across twelve behaviours the union reaches 4 (99%) / 2 (90%) of 7, leaving a residual 2–3 directions that no behaviour constrains. Four genuinely distinct locomotor presets (gentle/harsh $\times$ anterior/posterior touch) each constrain 2–4 of 7 mechanism directions, their leading stiff subspaces only partially overlapping (mean pairwise $\cos 0.54$, range 0.35–0.77), and their union is 5 (99%) / 2 (90%). Quantifying the complementarity directly: the four behaviour-specific stiff directions span an effective 2.98 of 4 dimensions (a single shared

direction would give 1; the participation dimension of their singular values), and each behaviour contributes a mean 0.535 component orthogonal to the span of the other three (per-behaviour 0.56/0.65/0.46/0.47). To establish that this growth is not the trivial consequence of non-identical subspaces, we compare the observed geometry against two explicit nulls, with 20,000 draws each. Under a *redundant* null, in which every behaviour constrains one shared direction, the span would be 1, the novel fraction 0 and the mean pairwise cosine 1. Under a *random-orientation* null, with directions drawn uniformly on the sphere in the same ambient space, the span is 3.50 (95% interval 3.00–3.88), the novel fraction 0.735 and the mean pairwise cosine 0.312. The observed values (span 2.978, novel fraction 0.535, cosine 0.525) lie between the two and are significantly more aligned than random ($p=0.020$ for the span, $p=0.082$ for the novel fraction, $p=0.016$ for the cosine). The behaviours therefore share substantial structure while each still contributes a distinct component. With only four behaviours, a bootstrap over behaviours is weak—resampling can draw duplicates and collapse the span—so we report the null comparison as the primary statistic and the bootstrap interval (span 1.45–2.98) only for completeness. File `null_model_tiling.json`. The residual core is genuine biological degeneracy, not an unused parameter (S6.7). Files `SAT_saturation.json`, `MB_behaviour_specificity.json`.

5.5 Cross-species: the same growth structure across three phyla (S6.5)

The same union construction applied to two further simulators shows the identifiable dimension scaling with behavioural richness in each. *Fly larva* (larvaworld; 10 behavioural-module parameters): four genuinely different sensory behaviours (free exploration, chemo-orbiting, anemotaxis, feeding) each have single-behaviour effective dimension 3–5 of 10 and engage different modules (olfactory, wind-sensory, feeding); their union is 8 (99%) / 4 (90%), roughly double the single-behaviour value. *Mammal* (Virtual Rodent (19, 20), `dm_control` (21), MuJoCo (22); 38 actuators): a six-behaviour repertoire drives a still-rising union curve ($3.67 \rightarrow 5.93 \rightarrow 7.67 \rightarrow 9.27 \rightarrow 10.5 \rightarrow 12.0$). The absolute counts are not comparable across species—the three models are parameterised differently and the rodent uses open-loop sinusoidal drives rather than trained natural behaviours—so we report the invariant structure (per-behaviour low-dimensional, complementary across behaviours, union set by repertoire), not a cross-species numerical law. Files `LARVA_manifold.json`, `RODENT_manifold.json`.

5.6 Concordance of two independently constructed worm models (S6.6)

Probed identically (per-mechanism Gauss–Newton elasticity, the same six observables, $\delta=0.25$), modWorm (coupled-ODE, 7 mechanisms) and BAAIWorm (NEURON, 20 named channels) agree at the class level: the network-wiring axes (synaptic, gap-junction) carry a stiff fraction of 0.833 (modWorm) / 0.868 (BAAIWorm), the single-cell biophysical axes only 0.083 / 0.158. “Wiring stiff, single-cell sloppy” is thus a property of the worm, not of one implementation. (The fine ordering *within* the stiff classes differs between the two models—modWorm ranks synaptic highest, BAAIWorm the passive leak—so the rank concordance over $n=3$ shared classes is not used as evidence; only the class-level split is.) File `G2_crosssim_align.json`.

5.7 Sloppy directions as coupled parameter combinations (S6.7)

On the real modWorm model the stiff and sloppy Hessian eigenvectors are both distributed parameter *combinations*: stiff-eigenvector participation 3.42, sloppy 4.68 of 7. Moving a fixed step along the sloppy eigenvector changes behaviour less than moving along the stiff eigenvector or along a single parameter (stiff/sloppy behavioural-change ratio 0.74; single-parameter/sloppy 0.97 at step

0.6), confirming the sloppy direction is a behaviour-preserving coupled combination rather than a parameter with no effect. The residual degenerate core is therefore a true biological neutral space. File `CP_coupling.json`.

5.8 Robustness of the effective-dimension measurement (S6.8)

Three controls establish that the low effective dimension is neither a single-point nor a method artefact. (i) *Multiple manifold points*: at five independent points on the real modWorm manifold the per-mechanism Hessian effective dimension is $[4, 3, 1, 3, 4]$ of 7—low everywhere, not at one optimum. (ii) *Ruler calibration*: on constructed spectra of known rank the estimator is exact (flat- k spectra recover $k=2, 5, 10$; uniform- d recover $d=8, 20$; an end-to-end sloppy geometric spectrum recovers 2). (iii) *Statistics of probe agreement*: the Hessian-stiffness versus manifold-drift ranking agree at Spearman $\rho=0.964$, exact permutation $p=0.0028$ (5040 permutations), bootstrap 95% CI $[0.565, 1.00]$. Files `mw_D1R6.json`, `D2_ruler_control.json`, `D3_modworm_stats.json`.

5.9 Robustness of the low dimension to the spectral-mass threshold (S6.9)

The effective dimension is the number of eigenvalues reaching a chosen fraction of the spectral mass; the main text reports the 90% and 99% values. Recomputing across thresholds 80/90/95/99%, and alongside the threshold-free participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$), the low-dimensional conclusion is insensitive to the choice (Table 2): every system stays low-dimensional at all thresholds, and the participation ratio—which uses no threshold—is ≤ 3.4 throughout. The rodent union is the highest, exactly as the behavioural-repertoire law predicts (a richer repertoire constrains more directions), not because of a threshold artefact.

system	dim	effective dimension at threshold				PR
		80%	90%	95%	99%	
modWorm full Hessian	30	1	1	1	1	1.00
modWorm 7-mechanism	7	2	3	3	4	2.23
modWorm rich-observable	7	1	2	2	3	1.35
BAAIWorm per-channel	20	1	2	2	3	1.34
larva union (4 behaviours)	10	3	4	5	8	2.07
rodent union (6 behaviours)	38	4	6	7	≥ 10	3.35

Table 2: Effective dimension is low across spectral-mass thresholds and by the threshold-free participation ratio (PR). The rodent row uses the stored top-10 union spectrum, so its 99% value is a lower bound (the full-spectrum union is 12, S6.5). Files as in S6.4–S6.8 and `modworm_hessian_full.json`.

6 Reproducibility

All external datasets are public: Prinz–Marder STG model population (7); Allen Cell Types; WormBase ParaSite / Ensembl Metazoa; OpenWorm Movement Database. All simulators are the published releases cited in the main text, run at the settings given above. Each result section names its saved result file; every reported quantity and figure is regenerated from those files by the accompanying reproduction code.

References

- [1] Balázs Szigeti, Padraig Gleeson, Michael Vella, et al. OpenWorm: An open-science approach to modeling *Caenorhabditis elegans*. *Frontiers in Computational Neuroscience*, 8:137, 2014.
- [2] Padraig Gleeson, David Lung, Radu Grober, et al. c302: a multiscale framework for modelling the nervous system of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170379, 2018.
- [3] George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [4] Alvaro Tejero-Cantero, Jan Boelts, Michael Deistler, Jan-Matthis Lueckmann, Conor Durkan, Pedro J Gonçalves, David S Greenberg, and Jakob H Macke. sbi: A toolkit for simulation-based inference. *Journal of Open Source Software*, 5(52):2505, 2020.
- [5] Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv preprint arXiv:1804.06788*, 2018.
- [6] Joeri Hermans, Arnaud Delaunoy, François Rozet, Antoine Wehenkel, Volodimir Begy, and Gilles Louppe. A trust crisis in simulation-based inference? Your posterior approximations can be unfaithful. In *Transactions on Machine Learning Research*, 2022.
- [7] Astrid A Prinz, Dirk Bucher, and Eve Marder. Similar network activity from disparate circuit parameters. *Nature Neuroscience*, 7(12):1345–1352, 2004.
- [8] Eve Marder and Adam L Taylor. Variability, compensation and homeostasis in neuron and network function. *Nature Reviews Neuroscience*, 12(10):591–603, 2011.
- [9] Mark S. Goldman, Jorge Golowasch, Eve Marder, and L. F. Abbott. Global structure, robustness, and modulation of neuronal models. *Journal of Neuroscience*, 21:5229–5238, 2001.
- [10] Adam L Taylor, Jean-Marc Goaillard, Davi Bhatt, and Eve Marder. How multiple conductances determine electrophysiological properties in a multicompartiment model. *Journal of Neuroscience*, 29(17):5573–5586, 2009.
- [11] Leandro M Alonso and Eve Marder. Visualization of currents in neural models with similar behavior and different conductance densities. *eLife*, 8:e42722, 2019.
- [12] Eve Marder and Jean-Marc Goaillard. Variability, compensation and homeostasis in neuron and network function. *Nature Reviews Neuroscience*, 7:563–574, 2006.
- [13] Michael L. Hines and Nicholas T. Carnevale. The NEURON simulation environment. *Neural Computation*, 9(6):1179–1209, 1997.
- [14] Saul Kato, Harris S Kaplan, Tina Schrödel, et al. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell*, 163(3):656–669, 2015.
- [15] Janne K. Lappalainen, Fabian D. Tschopp, Sridhama Prakhya, Mason McGill, Aljoscha Nern, Kazunori Shinomiya, Shin-ya Takemura, Eyal Gruntman, Jakob H. Macke, and Srinivas C. Turaga. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634:1132–1140, 2024.

- [16] Louis K. Scheffer, C. Shan Xu, Michal Januszewski, et al. A connectome and analysis of the adult *Drosophila* central brain. *eLife*, 9:e57443, 2020.
- [17] Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, et al. Neuronal wiring diagram of an adult brain. *Nature*, 634:124–138, 2024.
- [18] Greg J Stephens, Bethany Johnson-Kerner, William Bialek, and William S Ryu. Dimensionality and dynamics in the behavior of *C. elegans*. *PLoS Computational Biology*, 4(4):e1000028, 2008.
- [19] Josh Merel, Diego Aldarondo, Jesse Marshall, Yuval Tassa, Greg Wayne, and Bence Ölveczky. Deep neuroethology of a virtual rodent. *arXiv preprint arXiv:1911.09451*, 2019. International Conference on Learning Representations (ICLR) 2020.
- [20] Diego Aldarondo, Josh Merel, Jesse D. Marshall, Leonard Hasenclever, Ugne Klibaite, Amanda Gellis, Yuval Tassa, Greg Wayne, Matthew Botvinick, and Bence P. Ölveczky. A virtual rodent predicts the structure of neural activity across behaviours. *Nature*, 632(8025):594–602, 2024.
- [21] Saran Tunyasuvunakool, Alistair Muldal, Yotam Doron, Siqi Liu, Steven Bohez, Josh Merel, Tom Erez, Timothy Lillicrap, Nicolas Heess, and Yuval Tassa. dm_control: software and tasks for continuous control. *Software Impacts*, 6:100022, 2020.
- [22] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.